

SPC Visual Posters - Quality Classroom and Shop Floor

A3 reference posters | AIAG SPC 2nd Edition

QUALITY HEAD AGENTS | AIAG SPC 2nd Edition | IATF 16949:2016

POSTER 1: Control Limit vs Spec Limit - The Golden Rule

- CONTROL LIMITS (solid red lines) = What the process does. Calculated from data. $UCL = \bar{X} + 3\sigma$.
- SPEC LIMITS (dashed lines) = What the customer needs. From engineering drawing.
- NEVER set control limits equal to spec limits. This is the #1 SPC mistake.
- If a point is outside control limits: STOP. Investigate. Do not adjust randomly.
- If a point is inside control limits but outside spec: part is defective. Sort.

POSTER 2: 4 Key Nelson Rules - Know These Before You Plot

- Rule 1: ANY point outside UCL or LCL = STOP. Special cause. Act now.
- Rule 2: 9 points in a ROW on SAME side of centerline = Process has shifted. Investigate.
- Rule 3: 6 points TRENDING up or down = Tool wear, fixture drift, temperature. Act now.
- Rule 4: 14 points ALTERNATING up-down = Tampering. Stop adjusting - observe and call QA.
- KEY RULE: Never adjust the process just because of one high or one low reading inside limits.

POSTER 3: Cpk Verdict Card - Quick Reference

- $Cpk \geq 1.67$: EXCELLENT - 5 sigma. CC characteristic standard. No action needed.
- $Cpk 1.33 - 1.67$: ACCEPTABLE - 4 sigma. SC standard. Continue monitoring.
- $Cpk 1.00 - 1.33$: WARNING - 3 sigma. Process making ~2700 ppm defects. Improvement required.
- $Cpk < 1.00$: CRITICAL - Process is making non-conforming product. 100% inspection now.
- Remember: $Ppk \geq 1.67$ required for CC on PPAP submission. $Cpk \geq 1.67$ required ongoing.

POSTER 4: SPC Before PPAP - Checklist

- ☐ SPC plan defined for all CC/SC characteristics in Control Plan
- ☐ Control chart type selected: $\bar{X}$ -bar & R for variable, p-chart for attribute
- ☐ GRR < 30% confirmed for gauge used in SPC (ideally < 10% for CC)
- ☐ Operators trained on chart plotting and Nelson rule response
- ☐ Initial process study: 30 pieces, Ppk calculated
- ☐ $Ppk \geq 1.67$ for CC, $Ppk \geq 1.33$ for SC
- ☐ Out-of-control reaction plan posted at workstation
- ☐ Control chart template at workstation, ready for SOP
- ☐ Initial Ppk included in PPAP Element 9 package